



Forces

Ch 5 HRW

Newton's 1st, 2nd and 3rd law

Specific forces

Force diagrams

Newtonian Mechanics

- Speeds of the interacting bodies is much less than the speed of light.
- Objects are in the size range from very small (almost sub-atomic) to very large (galaxies).
- Before Newton:
 - To keep an object moving, it was necessary to apply a force to maintain constant velocity.
 - Natural state = at rest.
- Friction was not taken into account

Newton's 1st Law

- **Newton's First Law:** If a body is at rest, or is moving with constant velocity, it will continue to do so if no force **NET** acts on it.
- The object continues to move with constant velocity (magnitude and direction)
- $\vec{v} = \vec{v}_0$
 $\Delta\vec{v} = \vec{v} - \vec{v}_0 = 0$
 $\vec{a} = \frac{d\vec{v}}{dt} = 0$
- There is no acceleration if: $\vec{F}_{net} = \vec{F}_1 + \vec{F}_2 + \dots = 0$
 $\therefore \vec{F}_{net} = \sum \vec{F} = 0$

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Force

- A force is measured by the acceleration it produces.
- Force is a vector quantity (Magnitude and direction)
- Read section 5-4
- More than one force $\Rightarrow$ resultant (net) force
- Inertial frames – one in which Newton's laws hold

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Mass

- A given force will produce different accelerations for different objects.
- Kick/throw a soccer ball and a bowling ball with the same force – what are the effects?
- The mass of a body is the characteristic that relates the force on the body to the resulting acceleration.

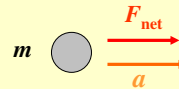
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Newton's 2nd Law

- **Newton's 2nd law:** The NET force on a body is equal to the product of the body's mass and its acceleration.

$$\vec{F}_{net} = m\vec{a}$$

- The direction of the net force is in the same direction as the acceleration
- Need to consider the NET force – i.e. the vector sum of all the forces acting on the body.
- Forces must act ON THAT BODY, not another body.



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Newton's 2nd Law

- Forces must act ON THAT BODY, not another body.
- Always consider

$$F_{net,x} = ma_x, F_{net,y} = ma_y, F_{net,z} = ma_z$$

- If net force = 0 N, then ALL components must also = 0 N i.e.

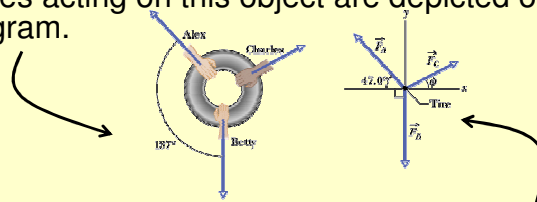
$$\text{If } \vec{F}_{net} = \sum \vec{F} = 0$$

$$\text{Then } F_{net,x} = ma_x = 0, F_{net,y} = ma_y = 0, F_{net,z} = ma_z = 0$$

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Newton's 2nd Law

- Equilibrium** – when the forces acting on a body balance each other out and there is no net force.
- Force diagram**: only one object is shown, and all the forces acting on this object are depicted on the force diagram.



- Free body diagram**: object is a point (representing the centre of mass) and the forces are drawn from the origin of the co-ordinate system.
- External force**: in a system of 1 or more bodies, any force from outside the system on the bodies is an external force.
- Internal force**: we do not include these in the FB diag.

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Checkpoint 2

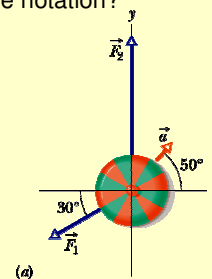
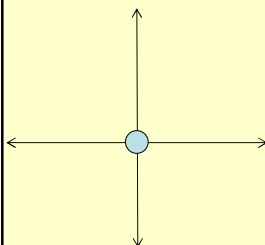
- The figure here shows two horizontal forces acting on a block on a frictionless floor. If a third horizontal force $\vec{F}_3$ also acts on the block, what are the magnitude and direction of $\vec{F}_3$ when the block is (a) stationary and (b) moving to the left with a constant speed of 5 m/s?



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Sample Problem p94

- In the overhead view of a 2.0 kg box is accelerated at 3.0 m/s^2 in the direction shown by $\vec{a}$, over a frictionless horizontal surface. The acceleration is caused by three horizontal forces, only two of which are shown: $\vec{F}_1$ of magnitude 10 N and $\vec{F}_2$ of magnitude 20 N. What is the third force $\vec{F}_3$ in unit-vector notation and in magnitude-angle notation?



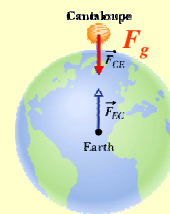
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Sample Problem p94

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Some Particular Forces

- **Gravitational Force:** It is the force that the Earth exerts on any object (in the picture a cantaloupe). It is directed toward the center of the Earth. Its magnitude is given by Newton's second law.



$$F_{net,y} = ma_y$$

$$-F_g = m(-g)$$

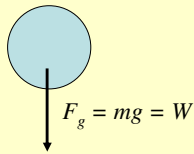
$$F_g = mg$$

$$\vec{F}_g = -F_g \hat{j} = -mg \hat{j}$$

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Some Particular Forces

- **Weight:** $\neq$ mass
- **Weight (W):** The weight of a body is equal to the magnitude F_g of the gravitational force on the body.
 - If there are two objects, suppose the gravitational force on the one is 2.0 N and the other one is 3.0 N. Then the second one is said to be heavier than the first.

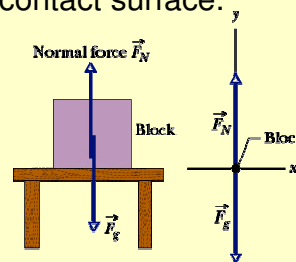


- If the object is moved to a location where the acceleration of gravity is different (e.g., the moon, where $g_m = 1.7 \text{ m/s}^2$), the mass does not change but the weight does.

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Some Particular Forces

- **Normal Force:** When a body presses against a surface, the surface deforms and pushes on the body with a normal force perpendicular to the contact surface.
- Symbol is: $\vec{F}_N$ or $\vec{N}$
- It is always perpendicular to the surface on which the object lies.
- Example – only 2 forces experienced by the block – no acceleration
- In this case: $F_{net,y} = F_N - F_g = ma_y$



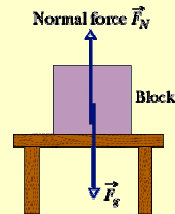
$$F_N = mg + ma_y = m(g + a_y)$$

$$F_N = mg$$

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Checkpoint 3

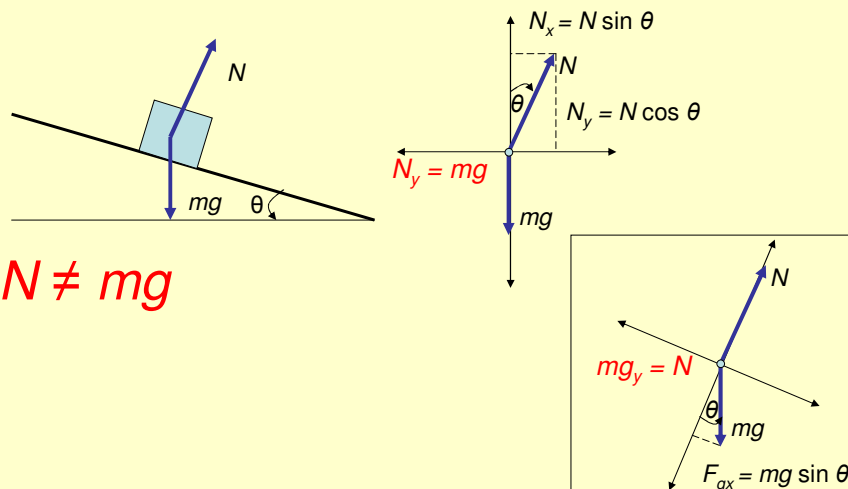
- Is the magnitude of the normal force greater, less than or equal to mg if the block and table are in a lift moving upward
 - at constant speed?
 - at increasing speed?



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Some Particular forces

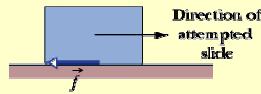
- Normal force – stationary object on a slope



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Some Particular Forces

- **Friction:** If we slide or attempt to slide an object over a surface, the motion is resisted by a bonding between the object and the surface. This force is known as “friction.”
- Symbol usually: $\vec{f}$
- Friction is always in the opposite direction to motion

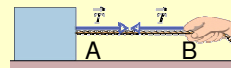


- Many problems state that motion is on a frictionless surface.

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Some Particular Forces

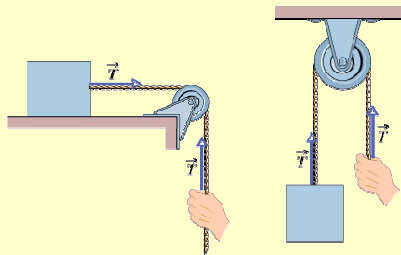
- **Tension:** This is the force exerted by a rope or a cable attached to an object
- Tension has the following characteristics:
 - It is always directed along the rope.
 - It is always pulling the object.
 - It has the same value along the rope (for example, points *A* and *B*).
- The following assumptions are made:
 - The rope has NEGLIGIBLE MASS compared to the mass of the object it pulls.
 - The rope does NOT STRETCH.



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Some Particular Forces

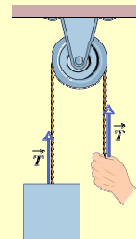
- **Tension:** This is the force exerted by a rope or a cable attached to an object
- If a pulley is used as in the figure, we assume that the pulley is massless and frictionless.
- Remember the tension is the same along the length of the rope.



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Checkpoint 4

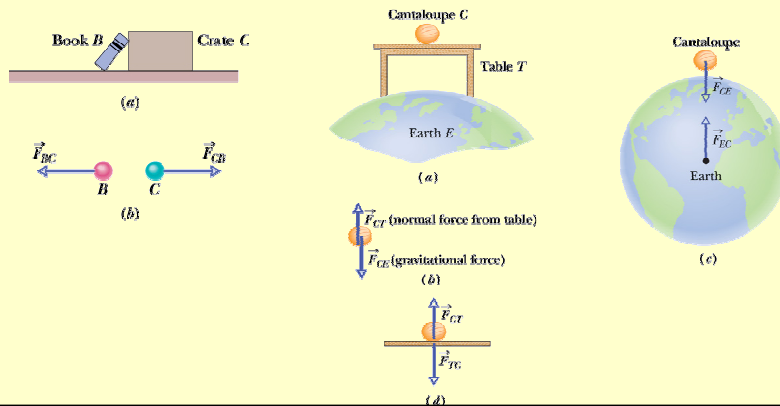
- The suspended body in the figure weighs 75 N. Is T equal to, greater than or less than 75 N when the body is moving upward
 - (a) at constant speed,
 - (b) at increasing speed, and
 - (c) at decreasing speed?



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Newton's 3rd Law

- **Newton's 3rd Law:** When two bodies interact by exerting forces on each other, the forces are equal in magnitude and opposite in direction.



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NB NOTE

- The rest of this chapter consists of sample problems. You should make sure you understand ALL of them, learning their procedures for attacking a problem. Especially important is knowing how to **translate a sketch** of a situation **into a free-body diagram** with appropriate axes, so that Newton's 2nd law can be applied.

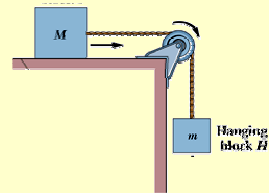
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Sample Problem p100

Read again carefully in HRW

- The Figure shows a block S (the sliding block) with mass $M = 3.3$ kg. The block is free to move along a horizontal frictionless surface and connected, by a cord that wraps over a frictionless pulley, to a second block H (the hanging block), with mass $m = 2.1$ kg. The cord and pulley have negligible masses compared to the blocks (they are “massless”). The hanging block H falls as the sliding block S accelerates to the right. Find (a) the acceleration of block S , (b) the acceleration of block H , and (c) the tension in the cord.
- What is the problem about?

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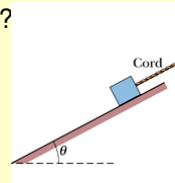
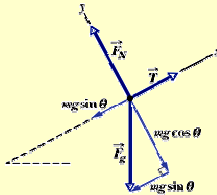
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Sample Problem p100

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Sample Problem p102

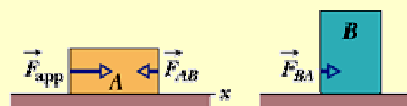
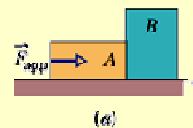
- In Fig. 5-16a, a cord pulls on a box up along a frictionless plane inclined at $\theta = 30^\circ$. The box has mass $m = 5.00$ kg, and the force from the cord has magnitude $T = 25.0$ N. What is the box's acceleration component a along the inclined plane?



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Sample Problem p104

- In the Figure, a constant horizontal force of magnitude 20 N is applied to block A of mass $m_A = 4.0$ kg, which pushes against block B of mass $m_B = 6.0$ kg. The blocks slide over a frictionless surface, along an x axis. (a) What is the acceleration of the blocks? (b) What is the (horizontal) force on block B from block A?



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Problem 53 (51)

- In the figure, three connected blocks are pulled to the right on a horizontal frictionless table by a force of magnitude $T_3 = 65.0$ N. If $m_1 = 12$ kg, $m_2 = 24.0$ kg and $m_3 = 31.0$ kg, calculate (a) the magnitude of the system's acceleration, (b) the tension T_1 and (c) the tension T_2 .

